

Dictionaries

TIP101

Agenda

1 Welcome 🙌

10 minutes

2 Time Management

10 minutes

3 Dictionaries

25 minutes

4 **BREAK**

5 minutes

5 AI Checkpoint #1

10 minutes

6 Collaborative Problem Solving

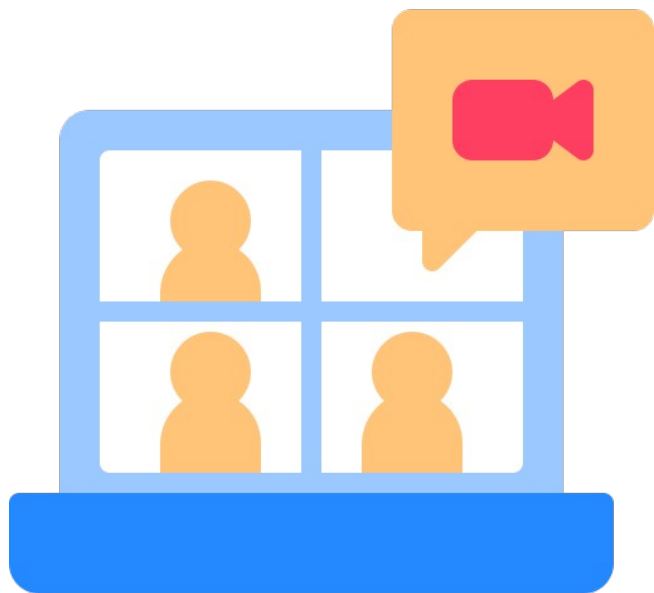
50 minutes

7 Wrap Up

10 minutes

10 minutes

Welcome! 🖐️



**Turn on your
cameras**



**Fill out the entry
ticket**



Icebreaker

5 minutes



Question:

Cooking is one of the best skills to have. Where do you rank yourself on the chef scale? What is your go to dish to make for an event?



Answer:

Share your answer in the Zoom chat below. See who your cooking competition might be!

Your Feedback



Glows

- + Students are excited to meet their classmates.
- + High class confidence level.



Grows

- VS Code logistics (Garage Session reminder!)
- Pacing discussions (unit test based pacing).

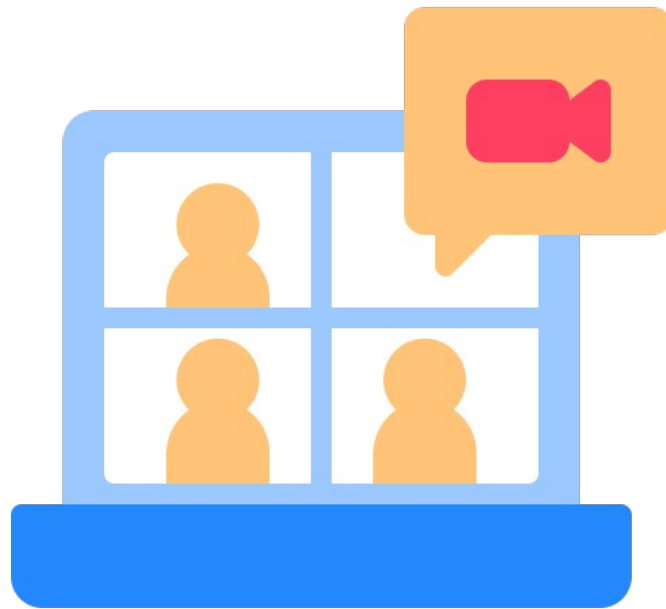


Announcements & Reminders

Don't Forget Cameras On!

You will be removed from today's session if you are not on camera and are unresponsive.

You will be unable to return to today's session if you are removed.





Announcements & Reminders

- * Class and study hall recordings have begun getting posted below.
- * They be found on the course portal under the Resources tab.
 - [Instructor Led Sessions Playlist](#)
 - [Study Hall Playlist](#)
 - [Fix-it Garage Playlist](#)

- * Assessment Reminder:
 - Students **must submit one** attempted HackerRank assessment **weekly** to remain in the course (due Sunday at 11:59pm PST). Please reach out to support channels if unexpected delays occur.
 - If a student moves ahead early in the units, and finishes passes all 10 units, they no longer require weekly attempts, but attendance is still mandatory.

Choosing Your Breakout Room

1. We'll share the link to the entry ticket so you can indicate what unit you are on.
2. We'll create breakout groups based on the unit you selected.
3. When it's Collaborative Problem Solving time, we'll share the room assignments and demonstrate how to get to your room.



10 minutes

Time Management



Try It!

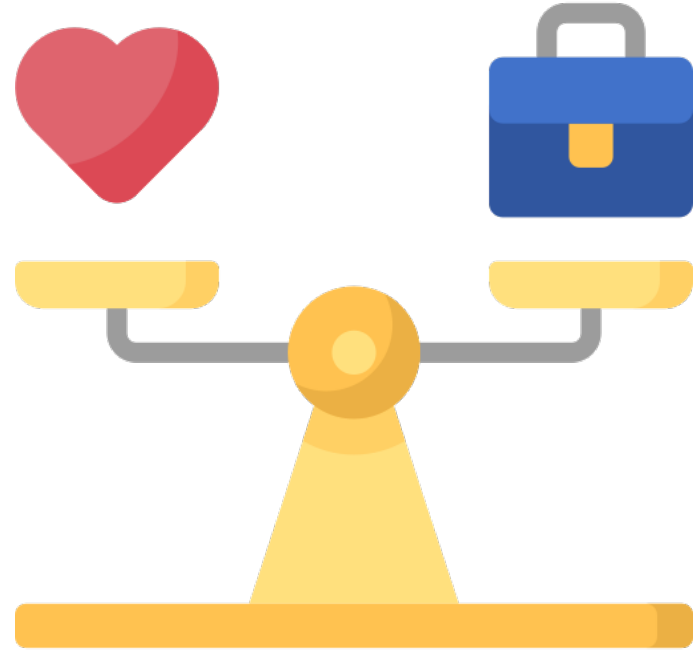
3 minutes

Grab a scrap piece of paper or a sticky note!

Write down **one short-term goal** for this week and **one long-term goal** for the course.

Balancing Study with Other Commitments

- * Identify non-negotiables (e.g., work, family) and fit study time around them.
- * Be realistic about what you can accomplish!





Discussion

3 minutes

In the chat . . .

What challenges do you face in balancing study with other commitments?

How do you plan to overcome them? Any suggestions for time management that work well for you? Think about how to integrate UMPIRE into your study approach. (Look up Leetcode Patterns)

25 minutes

Dictionaries



Discussion

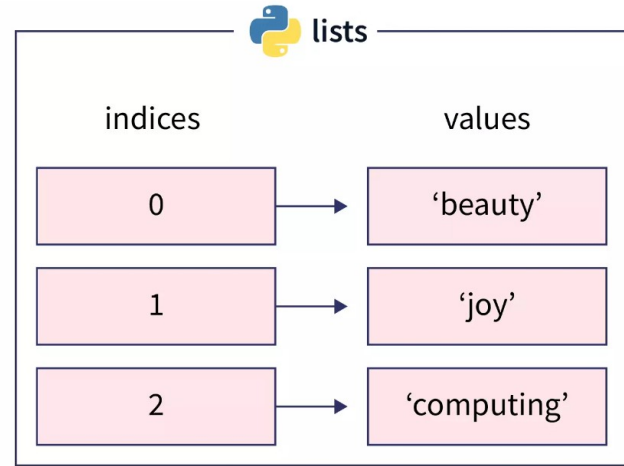
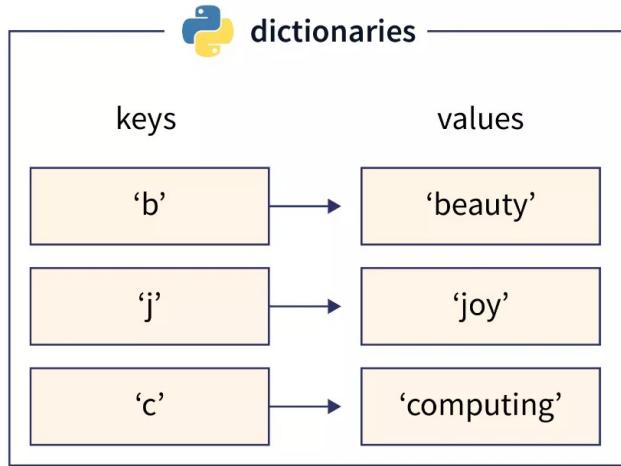
3 minutes

What do we know about dictionaries?



Dictionaries

- * Key-value pairs for fast lookups.
- * **Use cases:** Efficient retrieval and updating of data by unique keys.





Discussion

5 minutes

Open the [Unit 2 Cheat Sheet](#) and find **Dictionary Methods & Syntax**. Take a moment to review the functions and examples.

What do you notice?

What do you wonder?

Dictionary Methods & Syntax

Setting and Updating Values

You can add new key-value pairs or update the value of an existing key using the assignment operator `=`. If the key does not exist, a new key-value pair is added to the dictionary.

```
d['d'] = 4          # Adds a new key 'd' with value 4
print(d)           # Outputs: {'a': 1, 'b': 2, 'c': 3, 'd': 4}

d['a'] = 100        # Updates the value of key 'a'
print(d)           # Outputs: {'a': 100, 'b': 2, 'c': 3, 'd': 4}
```

Accessing Elements

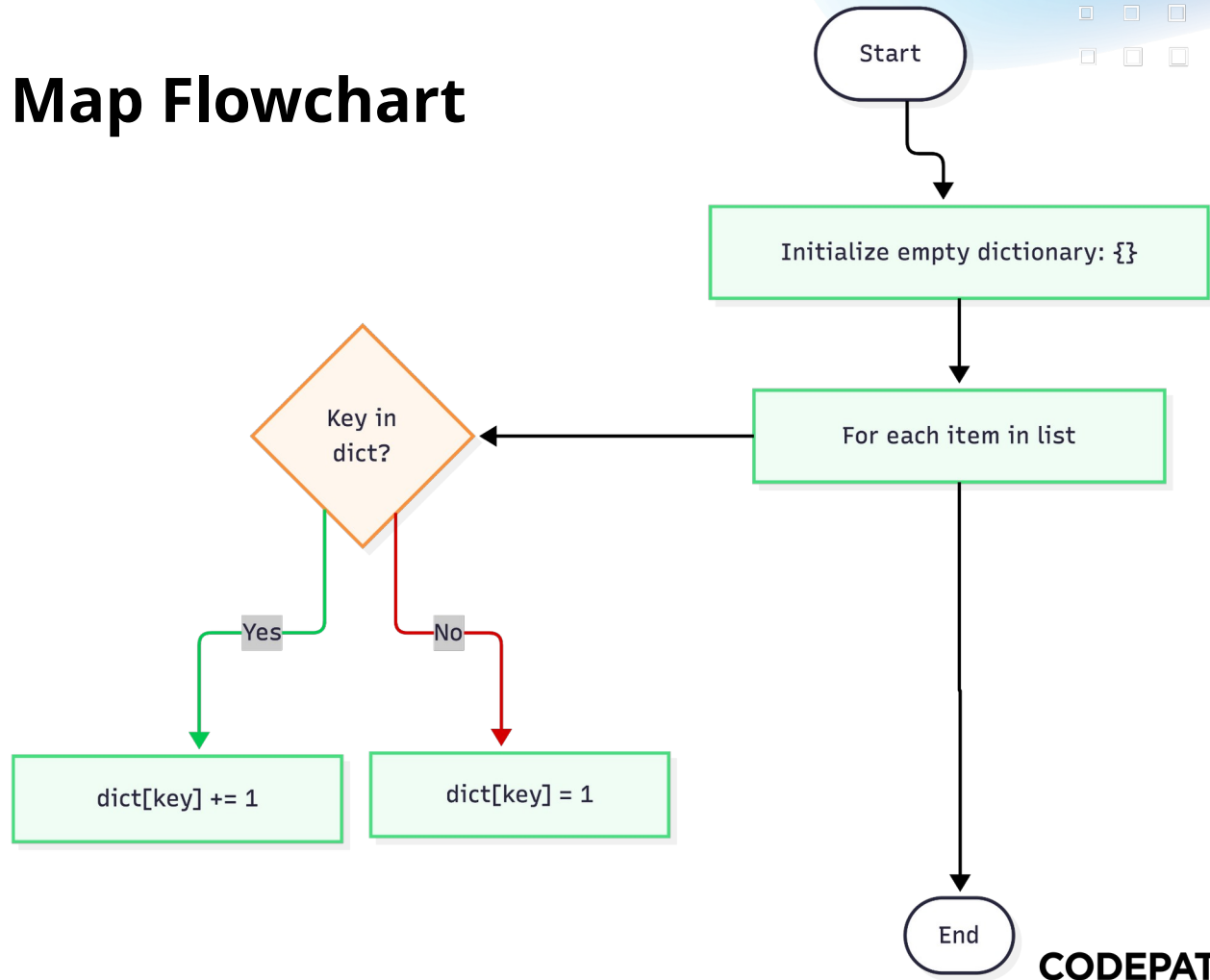
Dictionaries in Python allow you to access values using the keys. This can be done primarily in two ways:

Method 1: Direct access by Key

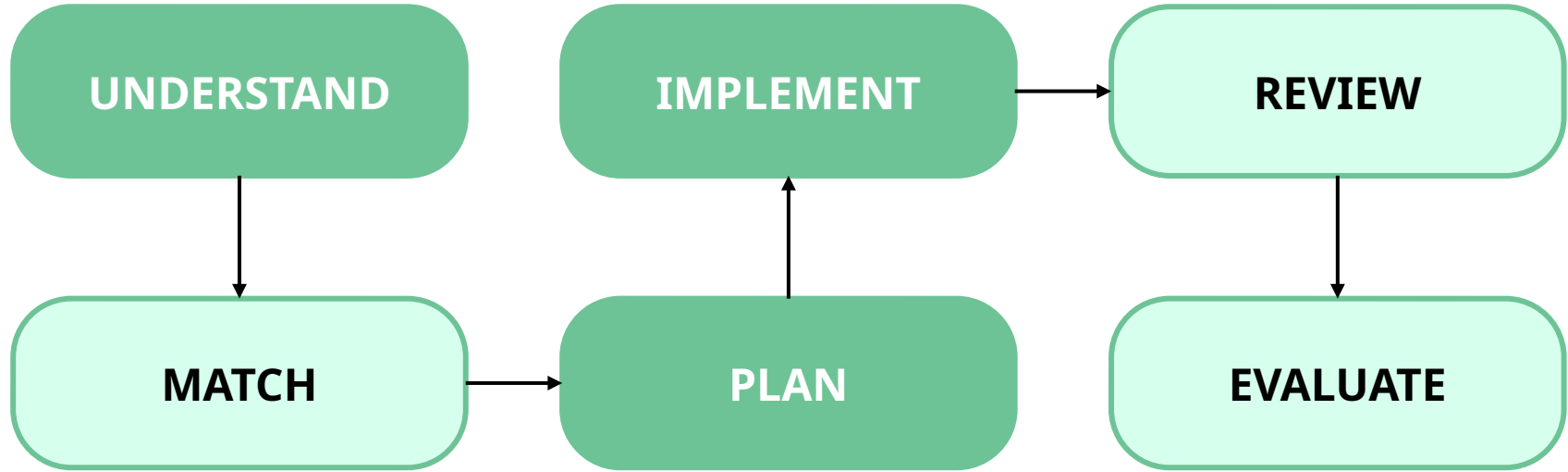
Common Use Cases

- * **Frequency maps:** Counting the frequency of characters in a string or elements in a list. This encompasses simple “**does it exist**” checks as well.
- * **Key-value data storage:** Storing data for quick 1 to 1 lookups:
 - Contact name : phone number
 - English word : Spanish word
 - Software setting : Is enabled? (yes or no)

Frequency Map Flowchart



UPI Recap





Instructor Demo

Given a list of strings **words**, group the strings that are anagrams of each other.

An anagram is defined as a word formed by rearranging the letters of another, such as **"eat"** and **"tea"**.

```
def group_anagrams(words):  
    pass
```

Input: words = ["eat", "tea", "tan", "ate", "nat", "bat"]

Output: [["eat", "tea", "ate"], ["tan", "nat"], ["bat"]]

Break Time!

Take 5 mins to step away from the computer. Feel free to turn off your camera and return promptly after 5 mins.



10 minutes

AI Checkpoint #1



Discussion

2-3 minutes

What **roles** can AI play in your learning?



Debugging
Partner



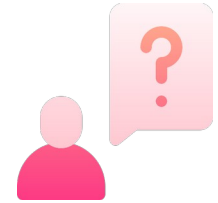
Virtual TA



Research
Assistant



CodePath
AI Tutor



What else?



Discussion

2-3 minutes

What's the **easiest mistake** to make when using AI?

What's the **best tip** you know on how to make better use of AI?



How to Use AI in this Course



Green Flags

Use AI to...

- + Clarify unfamiliar terms
- + Explain code concepts
- + Check your code for errors
- + Brainstorm ideas for your code



Red Flags

When using AI, don't...

- + Trust it blindly
- + Use AI-provided code without understanding it
- + Ignore best practices
- + Replace your own decision-making

50 minutes

Collaborative Problem Solving

Choosing Your Breakout Room

1. We'll share the room assignments.
2. We'll demonstrate how to find your room assignment and navigate to your room.





Breakout Rooms

50 minutes

1. 🏃 Navigate to your breakout room.
2. 🙌 Introduce yourselves!
3. 🧑💻 Work on the problems for your selected unit.

10 minutes

Wrap Up



One Good Thing

Let's reflect on today's session. Maybe you got a question answered, overcame a specific coding challenge, or learned how to do something completely new!

In the chat, share at least one good thing you've learned or experienced during the session today.

Let's celebrate together!



Session Survey Time!





Final Reminders

- ☐ Finish the problem sets for additional practice
- ☐ Take the weekly HackerRank assessment by Sunday @ midnight PST
- ☐ Visit a **Study Hall Session** to deepen your understanding, work through challenges, and connect with course material alongside your peers!